

BUKHARA INTL, Uzbekistan

WMO: 381415

Lat: 39.775N

Lon: 64.483E

Elev: 229

StdP: 98.61

Time Zone: 5.00 (E05)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-8.9	-17.2	0.8	-10.5	-13.9	1.2	-7.0	9.3	-1.1	8.0	3.7	2.9	340	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	40.2	19.6	39.0	19.0	37.8	18.5	21.1	37.1	20.2	35.8	19.5	34.7	5.1	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.2	11.1	25.1	14.2	10.4	24.8	13.8	10.1	24.7	61.6	36.7	58.4	36.3	56.2	34.9	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.0	7.2	DB	-13.2	43.0	4.0	1.6	-16.1	44.1	-18.5	45.0	-20.7	45.9	-23.7	47.0	(4)
(5)			WB	-13.6	22.7	4.1	1.0	-16.5	23.5	-18.9	24.1	-21.2	24.7	-24.1	25.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	2.1	4.0	11.4	17.3	23.8	28.2	29.7	27.6	22.2	15.1	8.3	3.3	(6)	
	DBStd	10.67	5.66	5.89	4.85	4.62	2.88	2.61	2.44	2.84	3.23	4.37	5.13	4.62	(7)	
	HDD10.0	778	249	176	42	4	0	0	0	0	0	10	87	211	(8)	
	HDD18.3	2087	503	401	219	72	1	0	0	0	6	117	303	465	(9)	
	CDD10.0	3025	4	9	86	223	427	547	612	545	365	167	37	4	(10)	
	CDD18.3	1292	0	0	4	41	169	297	354	287	121	16	2	0	(11)	
	CDH23.3	18032	0	1	81	554	2162	4320	5228	3889	1501	270	27	1	(12)	
	CDH26.7	10400	0	0	19	208	1076	2669	3330	2322	692	77	7	0	(13)	
(14) Wind	WSAvg	3.5	3.0	3.4	3.5	3.5	3.5	3.9	4.5	4.1	3.7	2.9	2.7	2.9	(14)	
(15) Precipitation	PrecAvg	145	19	24	28	22	11	2	1	1	1	3	16	17	(15)	
	PrecMax	206	49	63	48	60	53	11	5	2	5	14	63	38	(16)	
	PrecMin	74	5	6	6	1	0	0	0	0	0	0	1	2	(17)	
	PrecStd	40	11	14	13	16	13	3	1	1	2	4	14	12	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.9	21.2	29.1	35.0	37.8	41.2	43.0	41.0	37.0	31.9	27.1	19.2	(19)	
		MCWB	8.9	11.9	14.4	16.9	17.4	19.0	20.7	20.1	18.1	15.5	14.4	8.9	(20)	
	2%	DB	15.1	17.3	25.2	31.2	36.0	39.8	41.1	39.0	34.9	28.9	21.9	15.2	(21)	
		MCWB	8.0	10.2	13.5	15.9	17.1	18.8	20.1	19.3	17.0	14.2	11.8	8.1	(22)	
	5%	DB	12.2	14.2	22.9	29.0	34.2	38.1	39.8	37.8	33.0	26.2	18.9	12.8	(23)	
		MCWB	7.2	8.6	12.4	15.0	16.6	18.1	19.7	18.8	16.5	13.5	10.9	7.5	(24)	
	10%	DB	9.8	12.1	20.0	26.8	32.2	36.9	38.0	36.0	30.9	24.0	16.2	10.0	(25)	
		MCWB	6.1	7.8	11.7	14.3	16.4	17.8	18.8	18.0	15.8	13.0	9.6	6.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	12.7	16.0	18.7	19.5	20.8	22.5	22.3	19.1	16.7	14.7	10.4	(27)	
		MCDB	15.5	19.0	23.4	28.0	31.5	36.2	39.2	37.0	33.3	28.3	23.7	16.7	(28)	
	2%	WB	8.8	11.0	14.6	17.2	18.7	20.0	21.4	20.7	18.0	15.4	13.1	8.8	(29)	
		MCDB	14.4	15.8	22.7	26.9	30.4	35.5	37.7	36.0	32.1	26.1	20.2	13.7	(30)	
	5%	WB	7.5	9.6	13.3	16.4	18.0	19.2	20.6	19.8	17.1	14.3	11.5	7.6	(31)	
		MCDB	11.6	13.7	20.7	25.6	29.7	34.6	36.9	34.8	30.9	24.2	17.3	11.6	(32)	
	10%	WB	6.0	8.1	12.2	15.4	17.3	18.6	19.8	19.0	16.3	13.4	10.1	6.4	(33)	
		MCDB	9.3	11.3	19.2	24.1	29.0	33.8	35.7	33.8	29.5	22.7	15.6	9.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.7	11.8	13.1	14.7	15.4	15.3	15.5	15.7	14.6	11.7	9.1	(35)	
		MCDBR	13.5	13.6	16.4	17.1	17.5	17.4	17.2	17.1	18.0	17.6	15.6	13.4	(36)	
	5% WB	MCWBR	7.9	7.7	6.7	5.8	5.4	5.5	6.5	6.7	7.0	7.5	7.7	7.8	(37)	
		MCWBR	12.2	11.6	14.4	13.9	14.7	15.6	16.1	16.0	16.8	15.5	13.3	12.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.381	0.421	0.491	0.521	0.516	0.509	0.474	0.443	0.423	0.464	0.440	0.383	(40)	
		taud	2.122	2.021	1.876	1.840	1.874	1.895	1.992	2.061	2.071	1.930	1.979	2.100	(41)	
	Ebn at Noon		746	768	753	763	777	783	807	822	810	708	662	704	(42)	
		Edh at Noon	111	142	182	201	198	195	175	159	147	153	128	106	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.97	2.92	4.13	5.58	6.86	7.77	7.69	7.02	5.73	3.96	2.49	1.75	(44)	
	RadStd		0.17	0.29	0.33	0.45	0.30	0.23	0.17	0.14	0.15	0.21	0.23	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	+0.32	N/A	-0.80	N/A	N/A	N/A	+60	

Nomenclature: See separate page